Filippos Filippitzis

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Education

Sept. 2016 – Dec. 2020 — Ph.D., Department of Mechanical and Civil Engineering, **California Institute of Technology** (GPA 4.1).

Sept. 2016 – June 2018 — Master of Science in Mechanical Engineering, **California Institute of Technology** (GPA 4.1).

Sept. 2011 – July 2016 — Diploma in Mechanical Engineering, **University of Thessaly** (GPA 9.23/10).

Current Employment

Jan. 2026 – Present — **Siemens Digital Industries Software.** Software Development Engineer.
Continued in the same role following Altair's acquisition in 2026.

- Designed and implemented core mesh and geometry algorithms for the HyperMesh FEA preprocessing solution.
- Collaborated within a cross-functional team of over 40 engineers, developers, and product managers to deliver production-grade features and improvements.

Among other projects:

- Led performance optimization of the mesh rebuild tool, achieving up to a 90% speedup for larger (number of elements) models.
- Generalized the 2D hole washer treatment framework to support arbitrary hole types, improving algorithm flexibility and reuse.
- Increased mesh generator regression test coverage by 50%, significantly strengthening the reliability and enabling the successful delivery of the mesh generator project.

Sept. 2024 – Present — **University of Thessaly**. Adjunct Professor.
Responsible for teaching the undergraduate courses of *Ordinary Differential Equations* and *Computational Dynamics of Engineering Systems*, in the Department of Mechanical Engineering.

Work / Research Experience

Sept. 2022 – Jan. 2026 — **Altair Engineering.** Software Development Engineer. Design and implementation of mesh and geometry algorithms for the HyperMesh FEA preprocessing solution. *Role continued under Siemens Digital Industries Software following Altair's acquisition by Siemens in 2026.*

Aug. 2022 — **University of Thessaly.** Postdoctoral researcher in the System Dynamics Laboratory. Optimal sensor placement for response reconstruction and monitoring of structures.

Nov. 2021 – Aug 2022 — **Hellenic Air Force.** Military Service at the 111th Combat Wing, Nea Aghialos Air Base. Provided network administration, system maintenance, and technical support in a mission-critical military environment.

Oct. 2021 — **University of Thessaly.** Postdoctoral researcher in the System Dynamics Laboratory. Monitoring and response reconstruction in bridges using on-board measurements from passing trains, and application of sparse Bayesian learning techniques for leakage detection in water distribution networks.

Aug. 2014 – Sept. 2014 — **ETH Zurich.** Internship in the Computational Science and Engineering Laboratory. Exploring Bayesian Inversion Methods for Parameter Estimation of Models of Objects Falling in a Medium.

Areas or Interest / Expertise

Structural health monitoring; Digital twins; Model updating and parameter estimation; Bayesian learning; Mesh generation; Finite element modeling; Structural dynamics; Earthquake ground motion response; Earthquake data processing; Data analysis and visualization.

Teaching Experience

Feb. 2025 – Present	**University of Thessaly**, Department of Mechanical Engineering. **Lecturer**. *Computational Dynamics of Engineering Systems (MY3301)*, part of the undergraduate studies program.
Sept. 2024 – Present	**University of Thessaly**, Department of Mechanical Engineering. **Lecturer**. *Ordinary Differential Equations (ΓΕ0103)*, part of the undergraduate studies program.
Mar. 2021 – July 2021	**University of Western Macedonia**, Department of Mechanical Engineering. **Lecturer**. *Stress Analysis Methods: Theory, Simulation, Experiment (ADMES23)*, part of the graduate studies program in "Advanced Engineering of Energy Systems" (ADVENS).
Winter 2019	**California Institute of Technology**, Department of Mechanical and Civil Engineering. Teaching Assistant (TA). *Mechanics of Structures and Solids (Ae/AM/CE/ME 102B)*. TA to Professor John Hall.
Winter 2018	**California Institute of Technology**, Department of Mechanical and Civil Engineering. Teaching Assistant. *Experiments and Modeling in Mechanical Engineering (ME 050A)*. TA to instructor Dr. Michael Mello.

Ph.D. Research Work

My Ph.D. research focused on the area of structural health monitoring (SHM). I developed a structural damage identification methodology and software that allows detection, localization, and quantification of damage in a structure following a potentially damaging event. The method is based on sparse Bayesian Learning and integrates information from nonlinear finite element (FEM) models with information contained in dense accelerometer measurements. The developed approach enables model updating and model parameter estimation, combining physics-based simulation with measurement data in order to assess structural integrity under potentially damaging events. *(Journal publication [2]).*

As a member of Caltech's Community Seismic Network (CSN – csn.caltech.edu), I contributed to the processing, analysis, and visualization of large-scale seismic datasets collected from dense sensor networks in the Los Angeles region. A key component of this project included a collaborative effort with the NASA Jet Propulsion Laboratory (JPL) for dense accelerometer sensor deployment. The CSN sensor deployment achieved unprecedented density on the JPL campus in Pasadena, with an average inter-station spacing of approximately 200 m, enabling high-resolution analysis of ground-motion variability and wave propagation effects. The collected data was used in order to investigate urban ground-motion response and structural system behavior, as well as for the evaluation of 3D finite difference simulations and ground motion prediction equations *(Journal Publications [3,4]).*

Technical Skills – Programming Languages

- C++ (Daily development of mesh and geometry algorithms for HyperMesh)
- MATLAB (Multiple projects and daily use during my Ph.D., including the development of Structural Health Monitoring and earthquake signal processing tools)
- Fortran (Finite element software for beam and shell discretization and analysis)
- Python (Deep neural network for image classification)
- Tcl (Scripting for OpenSees and test framework development for Hypermesh)
- Html/CSS/JavaScript (Hobby - Personal webpage)

Technical Skills – Finite Element Modeling

- HyperMesh - HyperView (Altair Engineering - Siemens DISW)
- ANSA - META (Beta CAE)

- Ansys Mechanical, Ansys CFD (ANSYS)
- SAP2000, ETABS (CSI)
- OpenSees (UC Berkeley - Open source)

Technical Skills – Other

- Computer-aided design (AutoCAD)
- Version control (Git, Perforce)
- Document authoring (Microsoft Office, LaTeX)
- Product management (Jira, Confluence)
- AI-assisted development (GitHub Copilot)
- Generative AI & LLM productivity tools (Microsoft Copilot, Claude)

Selected Honours/Awards

2017-2020	The Cecil and Sally Drinkward Graduate Fellowship, Department of Mechanical and Civil Engineering, California Institute of Technology.
2016-2017	The Allan Acosta Endowed Graduate Fellowship, Department of Mechanical and Civil Engineering, California Institute of Technology.
2011-2016	State Scholarships Foundation (IKY) Fellowship and award received for 5 years, for ranking 1st during my studies in the Mechanical Engineering Department, University of Thessaly.
Sept. 2011	State Scholarships Foundation (IKY) Fellowship and award for high ranking during the National - Panhellenic Examination for admission to Greek Universities.
Sept. 2011	Eurobank Fellowship – Award for ranking first in his Lyceum - Senior High School.

Memberships and Affiliations

- Earthquake Engineering Research Institute (EERI)
- Seismological Society of America (SSA)
- Technical Chamber of Greece (TEE)

Publications

Journal

[J1] Siu, H.M., **Filippitzis, F.**, Stoura, C.D., Papadimitriou, C., and Dimitrakopoulos, E.G., 2024. "Utilizing On-board Sensing of Passing Train Vehicles for Virtual Sensing of Bridges." *Engineering Structures*. ***DOI: 10.1016/j.engstruct.2024.118808***.

[J2] **Filippitzis, F.**, Kohler, M.D., Heaton, T.H., and Beck J.L., 2021. "Sparse Bayesian Learning for Damage Identification using Nonlinear Models: Application in Weld Fractures of Steel-Frame Buildings." *Structural Control and Health Monitoring*. ***DOI: 10.1002/stc.2870***.

[J3] **Filippitzis, F.**, Kohler, M.D., Heaton, T.H., W. Graves, R.W., Clayton, R.W., Guy, G., Bunn, J.J., and Chandy, K.M., 2021. "Ground Motion Response in Urban Los Angeles from the 2019 Ridgecrest Earthquake Sequence." *Earthquake Spectra*. ***DOI: 10.1177/87552930211003916***.

[J4] Kohler, M.D., **Filippitzis, F.**, Heaton, T.H., R.W., Clayton, R.W., Guy, G., Bunn, J.J., and Chandy, K.M., 2020. "2019 Ridgecrest Earthquake Reveals Areas of Los Angeles That Amplify Shaking of High-Rises." *Seismological Research Letters*. ***DOI: 10.1785/0220200170***.

[J5] **Filippitzis, F.**, Gourgoulianis, K., Daniil, Z. and Bontozoglou, V., 2020. "The Effect of Alveolar Mixing on Particle Retention and Deposition Investigated by a Dynamic Single-Path Model." *Aerosol Science and Technology*. ***DOI: 10.1080/02786826.2020.1759775***.

Conference Proceedings

[C1] **Filippitzis, F.**, Siu, H.M., Stoura, C.D., Papadimitriou, C., and Dimitrakopoulos, E.G., 2023. "Response Reconstruction in Bridges Using On-board Measurements from Passing Vehicles." *5th ECCOMAS Thematic Conference on Uncertainty Quantification in Computational Science and Engineering (UNCECOMP 2023), ECCOMAS Proceedia*. ***DOI: 10.7712/120223.10322.20017***. ***Proceedings paper***.

[C2] Kohler, M.D., **Filippitzis, F.**, Graves, R.W., Massari, A.T., Heaton, T.H., Clayton, R.W., Bunn, J.J., Guy, R., and Chandy, K.M., 2022. "Variations in Ground Motion Amplification in the Los Angeles Basin due to the 2019 M7.1 Ridgecrest Earthquake: Implications for the Long-Period Response of Infrastructure." *ASCE Lifelines 2022.* ***DOI: 10.1061/9780784484449.020***. ***Proceedings paper***.

[C3] Ercan, T., **Filippitzis, F.**, and Papadimitriou, C., 2022. "Optimal Design of Sensor Networks for Monitoring of Structures." 5th Hellenic Conference on Earthquake Engineering and Engineering Seismology (HAEE/ETAM), 20-22 Oct. 2022, Athens, Greece. ***Proceedings paper***.

[C4] **Filippitzis, F.**, Kohler, M.D., Massari, A.T., Roh, B., and Heaton, T.H., 2021. "Spectral Scaling Transfer Function Method for Scenario Ground Motion Simulation with Application to the 2019 Ridgecrest Earthquake Sequence." *2021 SCEC Annual Meeting.* ***Poster Presentation***.

[C5] Kohler, M.D., Massari, A.T., **Filippitzis, F.**, Heaton, T.H., and Roh, B., 2021. "Spectral Scaling Transfer Function Method for Scenario Ground Motion Simulation with Application to the 2019 Ridgecrest Earthquake Sequence." *2021 AGU Fall Meeting.* ***Abstract***.

[C6] **Filippitzis, F.**, Kohler, M.D., Heaton, T.H., Graves, R.W., Clayton, R.W., Guy, R., Bunn, J.J., and Chandy, K.M., 2020. "High-Resolution Site Response Study of the Los Angeles Basin from the 2019 Ridgecrest Earthquake Sequence." 2021 *SSA Annual Meeting.* ***Abstract and Oral Presentation***.

[C7] **Filippitzis, F.**, Kohler, M.D., Heaton, T.H., Graves, R.W., Clayton, R.W., Guy, R., Bunn, J.J., and Chandy, K.M., 2020. "Ground Motion in Urban Los Angeles from the 2019 Ridgecrest Earthquakes: Recorded Versus Model-Predicted Response." 2020 *AGU Fall Meeting.* ***Abstract and Oral Presentation***.

[C8] **Filippitzis, F.**, Kohler, M.D., Heaton, T.H., Graves, R.W., Clayton, R.W., Guy, R., Bunn, J.J., and Chandy, K.M., 2020. "Ground Motion Response Study of Urban Los Angeles following the 2019 Ridgecrest Earthquake Sequence." *2020 SCEC Annual Meeting.* ***Poster Presentation***.

[C9] **Filippitzis, F.**, Kohler, M.D., Heaton, T.H., 2019. "Identification of Sparse Damage in Steel-Frame Buildings Using Dense Seismic Array Measurements." *Structural Health Monitoring 2019. Proceedings of the 12th IWSHM, Stanford.* ***DOI: 10.12783/shm2019/32398***. ***Proceedings paper and Oral Presentation***.

[C10] **Filippitzis, F.**, Kohler, M.D., Heaton, T.H., 2019. "Identification of Sparse Damage in Steel Buildings using Community Seismic Network." *ASCE-EMI (*Engineering Mechanics Institute*) 2019 Conference, Caltech.* ***Abstract and Oral Presentation***.

[C11] **Filippitzis, F.**, Kohler, M.D., Clayton, R.W., Bunn, J.J., Guy, G., Heaton, T.H., and Chandy, K.M., 2018. "Low-cost Seismic Monitoring: The Community Seismic Network." *2018 Geomechanics and Mitigation of Geohazards Fall Meeting (GMG 2018).* ***Poster Presentation***.

Theses

[T1] **Filippitzis, F.**, 2020. "Identification of Structural Damage, Ground Motion Response, and the Benefits of Dense Seismic Instrumentation." ***Doctoral Dissertation***. *California Institute of Technology, Pasadena, USA. (defended 30 September 2020)*

[T2] **Filippitzis, F.**, 2016. "An Euler Model for Particle Transport and Deposition in Pulmonary Flows." ***Diploma Thesis***. *University of Thessaly, Greece*.

Reviewer for Journals:

- "Structural Control and Health Monitoring", published by Wiley.
- "Engineering Structures", published by Elsevier.